is also important to remember that, even for normalised vectors, $|\psi\rangle$ and $e^{i\theta}|\psi\rangle$ represent the same state. The global phase $e^{i\theta}$ has no physical significance.

Because $\mathcal{H}$ is a complex vector space, for any pair of states $|\psi\rangle$ and $|\phi\rangle$, any linear combination $\alpha|\psi\rangle + \beta|\phi\rangle$ with $\alpha, \beta \in \mathcal{C}$ is also a state. This is known as the *superposition principle*. Note that, as opposed to a global phase, the relative phase in such a superposition *is* physically significant. That is, $\alpha|\psi\rangle + \beta|\phi\rangle$ and $\alpha|\psi\rangle + e^{i\theta}\beta|\phi\rangle$ are not the same state.

An orthornomal basis for $\mathcal{H}$ is a discrete set of vectors $\{|n\rangle\}_n$ with $\langle n|m\rangle = \delta_{nm}$ where δ_{nm} is the Kroenecker delta function. Any state can then be expressed as

$$|\psi\rangle = \sum_n \alpha_n |n\rangle. \tag{2.1}$$

with $\alpha_n \in \mathcal{C}$. For normalised states, $\sum_n |\alpha_n|^2 = 1$. Vectors and their conjugates (kets and bras) can be represented by column and row vectors in a given basis

$$|\psi\rangle \sim \begin{pmatrix} \alpha_0 \\ \alpha_1 \\ \vdots \end{pmatrix} \qquad \text{and} \qquad \langle\psi| \sim (\alpha_0^*, \alpha_1^*, \cdots). \tag{2.2}$$

In finite dimension, the span of $\{|n\rangle\}_n$ equals $\mathcal{H}$ and it is thus a basis in the usual sense of linear algebra. In infinite dimensions, the span of $\{|n\rangle\}_n$ is required to be dense in $\mathcal{H}$ (completeness). It is also sometimes possible, and useful, to instead express states in terms of a continuous set of vectors (although dealing with normalisation becomes more subtle).

2.1.2 Evolution and operators

Recall that an operator $\hat{A}$ on $\mathcal{H}$ is a linear map from vectors to vectors

$$\hat{A}(\alpha|\psi\rangle + \beta|\phi\rangle) = \alpha\hat{A}|\psi\rangle + \beta\hat{A}|\phi\rangle. \tag{2.3}$$

In these notes, we use symbols with hats ($\hat{\cdot}$) to indicate operators. The Hermitian conjugate (or adjoint) $\hat{A}^\dagger$ of an operator $\hat{A}$ is defined by

$$\langle\hat{A}^\dagger\phi|\psi\rangle = \langle\phi|\hat{A}\psi\rangle \tag{2.4}$$

for all vectors $|\psi\rangle$ and $|\phi\rangle$, where $|\hat{A}\psi\rangle$ denotes $\hat{A}|\psi\rangle$ and similarly for conjugate vectors. An operator is Hermitian (or self-adjoint) if $\hat{A}^\dagger = \hat{A}$ and it is unitary if $\hat{A}^\dagger\hat{A} = \hat{A}\hat{A}^\dagger = \mathbb{1}$, where $\mathbb{1}$ is the identity operator. An operator is positive semidefinite if $\langle\psi|\hat{A}|\psi\rangle \geq 0$ for all vectors $|\psi\rangle$. In a given orthonormal basis $\{|n\rangle\}_n$, any operator $\hat{A}$ can be represented by a matrix with elements $A_{mn} = \langle m|\hat{A}|n\rangle$. The Hermitian conjugate operator has matrix elements $(\hat{A}^\dagger)_{mn} = A^*_{nm}$, i.e. it corresponds to the transpose, conjugate matrix.

The time evolution of a closed quantum system is generated by a Hermitian operator $\hat{H}$ known as the Hamiltonian and is governed by the time-dependent Schrödinger equation

$$i\hbar\frac{d}{dt}|\psi(t)\rangle = \hat{H}|\psi(t)\rangle, \tag{2.5}$$

where $\hbar$ is the reduced Planck constant. It follows that evolution in closed quantum systems is always unitary, i.e.

$$|\psi(t)\rangle = \hat{U}(t)|\psi(0)\rangle \tag{2.6}$$